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ORIGINAL REPORT

Preliminary Transcriptomic Effects of Lateralized RLN Denervation in Rat Vocal Fold

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ABSTRACT

Objectives: To test whether unilateral recurrent laryngeal nerve (RLN) transection elicits side specific transcriptomic responses in rat vocal fold compartments. We compared the medial thyroarytenoid (MTA) muscle and mucosa on the left versus right, accounting for baseline laterality.

Methods: Unilateral RLN transection was performed in five adult rats (Left RLN: 2-males, 1-female; Right RLN: 1-male, 1-female). 5-weeks post-injury, the medial thyroarytenoid (MTA) muscle and vocal fold mucosa were harvested for RNA sequencing. Differential expression analysis compared Left-denervated versus Right-denervated tissues. A separate control Left-versus-Right analysis (1-male; 1-female) was used to assess baseline laterality and to contextualize TF transcription-factor inferences. TF activity was inferred using perturbation-based signatures, and baseline trends were used to classify overlapping signals as consistent with, opposite to, or distinct from baseline laterality.

Results: Left RLN denervation elicited distinct transcriptional responses in both compartments. In mucosa, TFs governing epithelial stability were reduced while regenerative cues were activated, suggesting impaired differentiation and recruitment of progenitor-like programs. In MTA, left-sided injury increased TFs consistent with stress signaling and mitochondrial compensation. TFs involved in epigenetic stability were reduced. Lateralized activity was observed in developmental regulators putatively related to positional memory or differential stress adaptation. Pathway analysis revealed enrichment of skeletal development processes in MTA and immune/matrix regulation in mucosa, indicating coordinated remodeling across muscle and epithelial compartments.

Conclusion: RLN injury produces side-specific muscle and mucosal changes, with left-sided denervation showing stronger atrophy and stress responses. These results emphasize the importance of laterality and coordinated tissue adaptation.

Level of Evidence: NA

1 | Introduction

Recurrent laryngeal nerve (RLN) injury commonly causes vocal fold (VF) paralysis (VFP), leading to dysphonia, impaired airway protection, and reduced quality of life [1–3]. The RLN innervates the intrinsic laryngeal muscles, including the thyroarytenoid (TA), and paralysis with atrophy is frequently observed after injury [4–6]. Unilateral VFP is more common on the left [7–9], likely reflecting the longer intrathoracic course of the left RLN. Anatomy alone, however, does not fully explain left-sided predominance. Left-biased cases persist in idiopathic presentations and after short intubations where mechanical factors are less compelling [10], and left–right differences in neurophysiology, myogenesis, inflammation, and

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hormonal signaling were reported in the intrinsic laryngeal muscles [11]. These observations suggest laterality may shape denervation responses. In addition, the TA may influence the overlying mucosa. TA myoblasts modulated vocal fold fibroblast gene expression, migration, and proliferation [12], raising the possibility that neuromuscular injury drives mucosal remodeling.

We investigated whether unilateral RLN transection elicits side specific transcriptomic changes in the medial TA (MTA) and VF mucosa. We hypothesized unilateral RLN transection would elicit side-dependent transcriptomic responses in both the MTA and mucosa, with greater remodeling after left-sided denervation. We further expected these responses to differ by tissue compartment, reflecting distinct patterns of post-injury adaptation in muscle and mucosa.

2 | Methods

All procedures were approved by the IACUC at the NYU Grossman School of Medicine (#202400004). Seven ($n = 7$) Sprague–Dawley rats (5-months-old) underwent left RLN transection (2 males, 1 female) or right RLN transection (1 male, 1 female). Two (1 male, 1 female) animals were anesthetized without nerve transection. The experimental unit was the individual rat. The primary outcome was differential gene expression between Left-denervated and Right-denervated tissue; the MTA and mucosa were analyzed separately. Secondary control analysis compared Left versus Right control tissue within each compartment to characterize baseline laterality.

This pilot study was conducted without an a priori power calculation. To limit false discovery with modest sample sizes, conservative fold-change thresholds (≥ 1.5 and ≥ 4) with FDR control ($q \leq 0.05$) were applied. Pathway-level and transcription factor signals were analyzed rather than single-gene effects. Sex was included as a covariate in DESeq2; only signals with consistent directionality across compartments were interpreted.

No animals or data were excluded. Allocation to side of transection was not randomized due to surgical logistics. Procedures were standardized and paired controls accounted for baseline left–right asymmetry. Samples were coded for RNA isolation, library preparation, sequencing, and bioinformatic analysis.

Rats were anesthetized with intraperitoneal ketamine (80 mg/kg) and xylazine (10 mg/kg) and maintained on 2% isoflurane. After sterile preparation, a midline cervical incision exposed the larynx and recurrent laryngeal nerve. An RLN segment (~8 mm) on the assigned side was excised between the third and fourth tracheal ring. Lidocaine (7 mg/kg) was administered, wounds were closed, and postoperative analgesia was provided (extended-release buprenorphine 0.65 mg/kg; meloxicam 1 mg/kg). Animals were euthanized 5 weeks after transection by CO_2 inhalation.

The larynx was removed and hemisected posteriorly to expose the vocal folds. Mucosa was separated from the thyroarytenoid as previously described [13]. Tissues were placed in cold PBS, minced and triturated, transferred to tubes, lysed in RLT buffer with 2-mercaptoethanol, and stored a-t −80°C. Left- and Right-denervated cohorts comprised independent animals; no within-animal contralateral comparisons were made. Accordingly, all denervation contrasts were between-animal Left-versus-Right comparisons conducted separately by tissue type (MTA or mucosa) using injured-side tissue only. Control animals provided paired Left and Right tissues for separate control analyses of baseline laterality. Control analysis was used only to contextualize transcription-factor perturbation results and was not reference tissue for the denervated differential-expression model. A schematic of the experimental paradigm is provided in Figure 1A,B.

2.1 | RNA Isolation and Library Preparation

Total RNA was isolated using the RNeasy Mini Kit (Qiagen) and RNA integrity was verified using an Agilent Bioanalyzer Pico Chip. Samples ≤ 1 ng were reverse transcribed using the SMART-Seq HT Kit (Takara) with 13 PCR cycles. cDNA was quantified using the Invitrogen Quant-iT system and diluted to 0.3 ng/μL for library preparation. Libraries were prepared with Nextera XT (Illumina), indexed, and quality-checked on an Agilent TapeStation. Libraries were pooled equimolarly and sequenced on an Illumina NovaSeq X+ platform (paired-end, 100 cycles).

2.2 | RNA Sequencing Quality Assessment

Sequencing quality was high (Q30, alignment) with acceptable library complexity (duplicates, multi-mapping); per sample metrics appear in- Table S1.

2.3 | RNA-Seq Processing and Differential Expression

Reads were trimmed with Cutadapt [14], quality-checked with FastQC [15], aligned to *Rattus norvegicus* BN7.2 using STAR [16], and counted with HTSeq [17]. Differential expression analysis was performed in Rosalind using DESeq2 [18] (RLE normalization) separately for each tissue (MTA and mucosa) and separately for denervated and control cohorts; sex was a covariate. Primary analysis compared Left-denervated versus Right-denervated samples within each tissue using injured-side tissues collected from different animals (between-animal comparison; Right-denervated as the reference level).

Separate control analyses compared left-control versus right-control tissues within each tissue type, with right-control specified as the reference level, to assess baseline laterality. These control-only comparisons were not included in the denervated DESeq2 analysis and were not used for pathway enrichment or volcano plots. Instead, the undenervated left-versus-right control comparisons were used only to contextualize transcription-factor perturbation results shown in Tables 1 and 2. This study did not test Left-injured versus Left-control or Right-injured versus Right-control contrasts.

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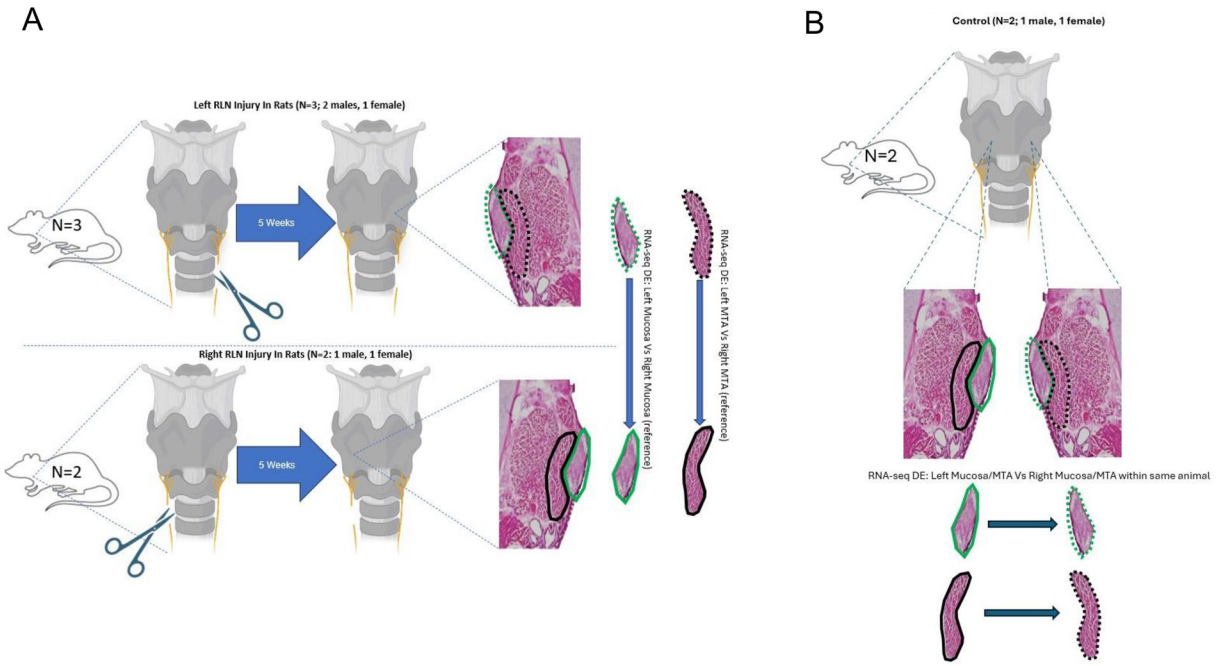


FIGURE 1 | Experimental design and analytical contrasts. (A) Unilateral RLN transection was performed in separate animals on either the left or right side. At 5 weeks post-injury, only denervated mucosa and MTA were collected, enabling a between-cohort comparison of Left-denervated versus Right-denervated samples within each tissue (Left: $N = 3$, 2 M/1F; Right: $N = 2$, 1 M/1F). (B) Sham animals provided paired left and right mucosa and MTA samples ($N = 2$, 1 M/1F) to assess baseline laterality in the absence of injury. Sham comparisons were used solely to contextualize intrinsic laterality in TF perturbation analyses and were not used for normalization or as reference in denervated differential expression.

No contralateral-based per-sample normalization was applied given the potential for contralateral remodeling after unilateral injury.

2.4 | Pathway and Transcription Factor Analysis

BioPlanet [19] enrichment was performed directly in Rosalind using the full set of differentially expressed genes for each tissue and contrast (Left vs. Right denervated) without manual or post hoc pathway selection/curation. Reported pathways passed FDR ($q \leq 0.05$). Rosalind automatically harmonized rat DEG lists to human orthologs for enrichment. Transcription factor activity was inferred using the TF Perturbations (CREEDS) library in Enrichr [20–22], applied separately to each upregulated and downregulated gene list (at |fold change| ≥ 4) and mapped to human orthologs using Ensembl [23]. Direction was inferred from alignment with overexpression or knockdown signatures; only predominant, significant ($q \leq 0.05$) results were reported. TFs were ranked by signed scores and $-\log_{10}(q)$.

3 | Results

3.1 | Global Differential Expression Overview

Genome-wide volcano plots summarizing primary denervated Left-versus-Right contrasts for MTA and mucosa are shown in Figure 2. Positive log2FC indicates higher expression in Left-denervated tissue; negative log2FC indicates higher expression in Right-denervated tissue. Vertical guide lines denote log2FC cutoffs corresponding to FC ≥ 1.5 and FC ≥ 4 and the horizontal threshold indicates $q = 0.05$.

3.2 | Pathway Enrichment Analysis

3.2.1 | Enrichment of Endochondral Ossification Pathway in Left-Denervated MTA

Endochondral ossification was significantly enriched in Left versus Right denervated MTA ($q = 0.03$). Contributing genes included SOX5, ADAMTS5, IGF1, HDAC4, and ADAMTS1 (upregulated on the Left) and SERPINH1, PTHLH, and PRKACA (downregulated on the Left). A heatmap summarizing genes contributing to this pathway across conditions is shown in Figure 3.

3.2.2 | Enrichment of Immune and ECM-Related Pathways in Left-Denervated Mucosa (|Fold Change| ≥ 1.5)

Two pathways were significantly enriched in mucosa—Interleukin-4 regulation of apoptosis ($q = 0.00068$) and TGF-β regulation of extracellular matrix ($q = 0.00078$). For Interleukin-4 regulation of apoptosis, most contributing genes were upregulated on the Left, including POSTN, VCAN, VCAM1, SELE, PTX3, DUSP6, GBP2, and LPL, while a smaller subset was downregulated, including MYC, MCM2, MCM3, and MCM5 (Figure 4A). For TGF-β regulation of extracellular matrix, the majority of contributors were upregulated on the Left, including C7, CCL2, VCAN, PDGFD, PDGFRA, AGTR1, VCAM1, IGF1, HAS2, LUM,

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TABLE 1 | TF activity (MTA).

	Denervated Left vs. Right			Control Left vs. Right		
TF	**Inferred direction**	***q***	**−log10 (*q*)**	**Inferred direction**	***q***	**−log10 (*q*)**
CEBPG	Decreased	6.3×10^{-3}	2.201		—	—
DUX4	Decreased	3.8×10^{-2}	1.421		—	—
EGR4	Increased	3.8×10^{-2}	1.421		—	—
ESRRA	Increased	3.8×10^{-2}	1.421		—	—
FOXA1	Decreased	3.8×10^{-2}	1.421	Increased	2.99×10^{-4}	3.524
FOXO1	Increased	6.3×10^{-3}	2.201		—	—
GATA2	Decreased	4.99×10^{-2}	1.302	Decreased	4.99×10^{-2}	1.302
GATA4	Increased	3.8×10^{-2}	1.421	Increased	2.8×10^{-2}	1.553
GATA6	Increased	3.8×10^{-2}	1.421	Mixed	↑ 1×10^{-2} | ↓ 2.57×10^{-2}	↑ 1.999 | ↓ 1.591
LIN28	Decreased	4.22×10^{-2}	1.374		—	—
MECP2	Decreased	3.8×10^{-2}	1.421	Increased	4.62×10^{-2}	1.335
NANOG	Increased	3.8×10^{-2}	1.421	Mixed	↑ 2.3×10^{-5} ↓ 1.3×10^{-2}	↑ 4.63 ↓ 1.87
NCOA2	Decreased	3.8×10^{-2}	1.421		—	—
NFE2L2	Mixed	↑ 3.8×10^{-2} | ↓ 3.8×10^{-2}	↑ 1.421 | ↓ 1.421	Mixed	↑ 7.25×10^{-3} ↓ 5.2×10^{-5}	↑ 2.140 | ↓ 4.284
NR2F2	Decreased	3.8×10^{-2}	1.421		—	—
PITX2	Increased	3.8×10^{-2}	1.421		—	—
RUNX1	Decreased	3.8×10^{-2}	1.421	Mixed	↑ 8.74×10^{-6} | ↓ 4.56×10^{-3}	↑ 5.059 | ↓ 2.341
SRF	Decreased	3.8×10^{-2}	1.421	Decreased	5.4×10^{-3}	2.268
U2AF1	Decreased	3.8×10^{-2}	1.421		—	—

Note: Inferred TF perturbation in Left versus Right comparisons for denervated and control cohorts using Enrichr CREEDS. Direction (increased/decreased) derived from enrichment in up−/downregulated DEG sets. Reports *q*-values and −log10 (*q*). "Mixed" indicates opposing directions across lists. DEGs: Rat (log2FC ≥ 2; $q \le 0.05$), mapped to human orthologs. Control columns reflect sham Left versus Right laterality only, not injury contrasts.

TABLE 2 | TF activity (mucosa).

	Denervated Left vs. Right			Control Left vs. Right		
TF	**Inferred direction**	***q***	**−log10 (*q*)**	**Inferred direction**	***q***	**−log10 (*q*)**
HES6	Increased	4.41×10^{-2}	1.355		—	—
IRF6	Decreased	1.4×10^{-2}	1.852	Increased	3.94×10^{-2}	1.404
NFE2L2	Mixed	↑ 1.4×10^{-2} | ↓ 1.4×10^{-2}	↑ 1.852 | ↓ 1.852	Mixed	↑ 1.6×10^{-5} | ↓ 6.4×10^{-5}	↑ 4.786 | ↓ 4.197
OVOL1	Decreased	1.4×10^{-2}	1.852	Increased	2.29×10^{-6}	5.639
OVOL2	Increased	1.4×10^{-2}	1.852	Mixed	↑ 5.94×10^{-3} | ↓ 7.93×10^{-3}	↑ 2.230 | ↓ 2.1
VAX2	Decreased	3.49×10^{-2}	1.457		—	—

Note: Same analytical framework as Table 1 using Enrichr CREEDS signatures and rat DEGs ($\log_2$FC ≥ 2; $q \le 0.05$) mapped to human orthologs. Control columns represent sham Left versus Right laterality only and are shown solely for baseline contextualization, not as injury reference.

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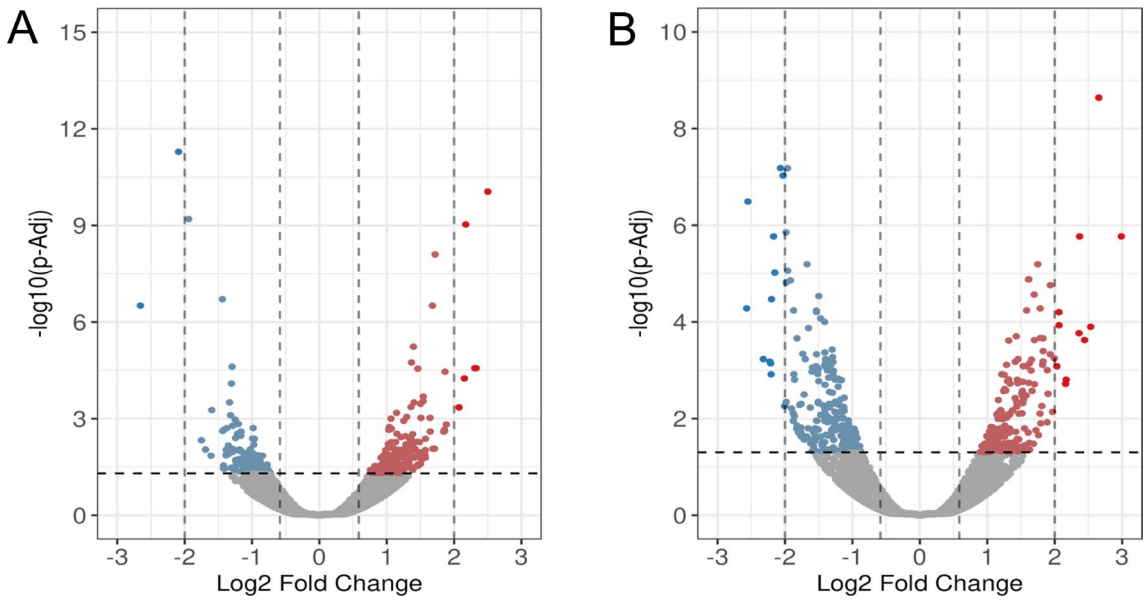


FIGURE 2 | Volcano plots (Left vs. Right denervated). Volcano plots for MTA (A) and mucosa (B). Positive log2FC indicates Left > Right. Thresholds shown: Log2FC = 0.585 (FC ≥ 1.5), log2FC = 2 (FC ≥ 4), and $q = 0.05$.

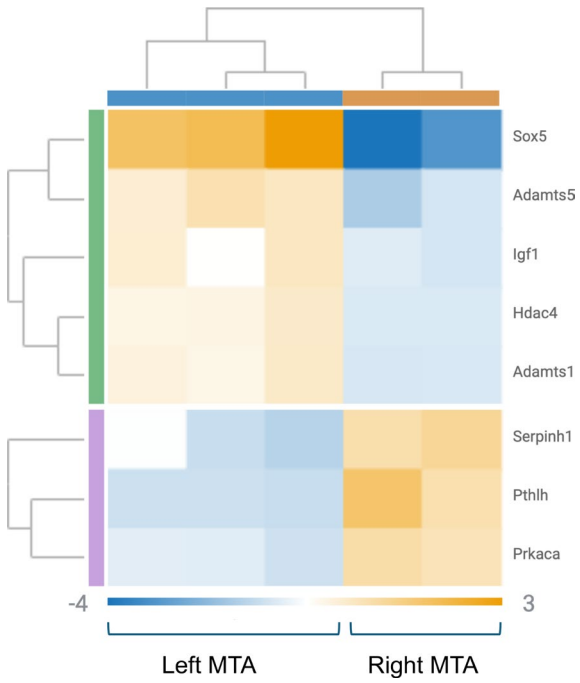


FIGURE 3 | MTA heatmap. Differential expression of endochondral ossification pathway genes (FC ≥ 1.5). Colors indicate relative expression (orange higher, blue lower). Pathways from BioPlanet; gene labels are rat symbols.

OGN, and ANGPT1, with a smaller set downregulated, including EPHA1, CST6, ELF3, MYC, MCM5, and JUP (Figure 4B).

3.2.3 | Enrichment of ECM and Cell-Communication Pathways in Left-Denervated Mucosa (|Fold Change| ≥ 4)

Two pathways remained significant at this stricter threshold—Small leucine-rich proteoglycan (SLRP) molecules ($q = 0.001$), with KERA and EPYC upregulated on the Left (Figure 5A), and Ephrin receptor A forward signaling ($q = 0.0173$), with EFNA3 and NGEF downregulated on the Left (Figure 5B).

3.3 | Transcription Factor Perturbation Analysis

Although DEG lists from control and denervated conditions did not directly overlap, several TFs were present in both contexts. For some, inferred activity was in the same direction in control and denervated comparisons, suggesting baseline laterality may persist after injury. Opposite directions were observed in other cases, indicating denervation responses differ from baseline patterns. Mixed control signatures were reported for completeness but interpreted cautiously. Importantly, all TFs in the denervation cohort were reported only when directionality was consistent within that context.

3.3.1 | Left-Sided Denervation Altered TFs Involved in Myogenesis, Mitochondrial Function, Stress Response, and Developmental Regulation

In the MTA, TFs with mixed regulatory signals in control comparisons included GATA6, NFE2L2, and RUNX1. TFs with consistent overlapping directionality between control and denervated conditions included GATA2, GATA4, and SRF, which may reflect baseline anatomical laterality rather than denervation-specific biology. TFs unique to denervation included DUX4, EGR4, ESRRA, FOXA1, LIN28, NANOG, NCOA2, NR2F2, PITX2, U2AF1, CEBPG, FOXO1, and MECP2 (Table 1).

3.3.2 | Suppression of Epithelial Stability and Activation of Regenerative Cues in the Left Mucosa

In VF mucosa, TFs with mixed regulatory signals in control comparisons included NFE2L2 and OVOL2. No TFs showed consistent overlapping directionality between control and denervated conditions. TFs unique to denervation included IRF6, OVOL1, VAX2, and HES6 (Table 2).

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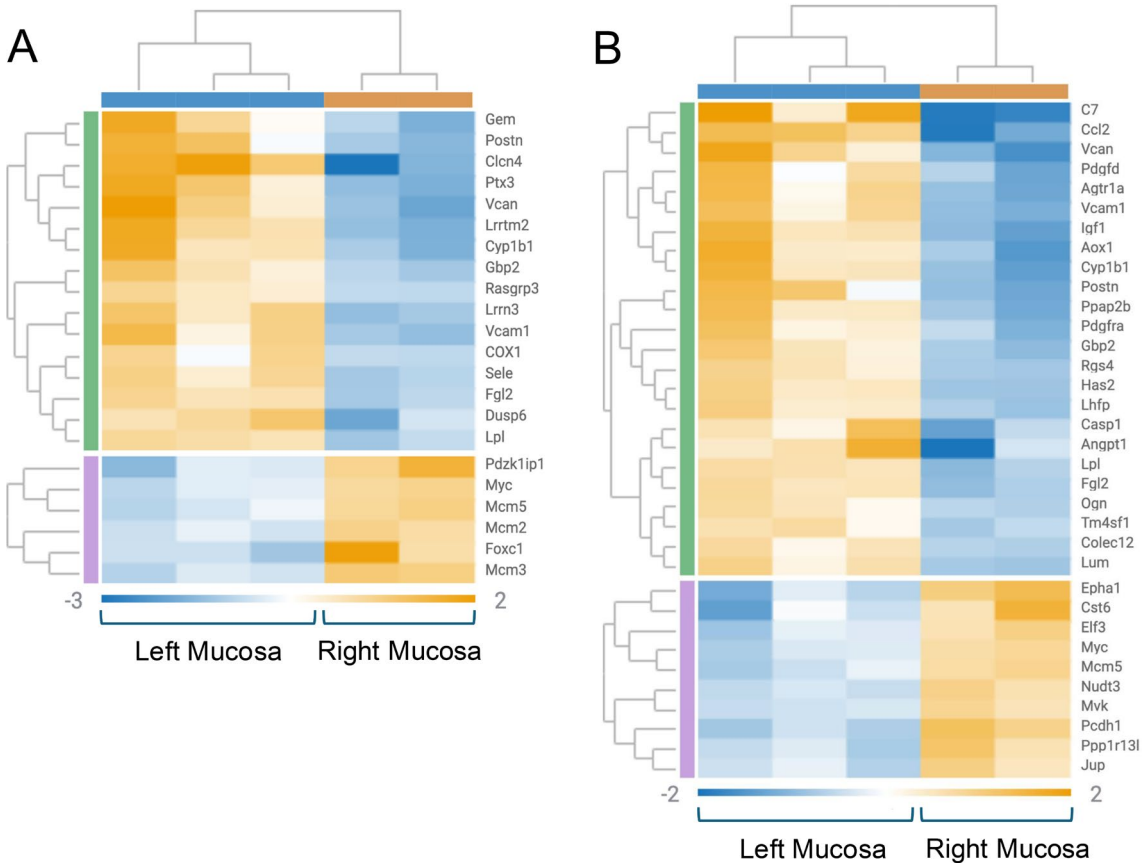


FIGURE 4 | Mucosa pathway heatmaps (FC ≥ 1.5). (A) IL4-mediated regulation of apoptosis. (B) TGFβ-mediated extracellular matrix regulation. Left-biased expression observed overall; cell cycle genes (e.g., MYC, MCMs) show reduced expression.

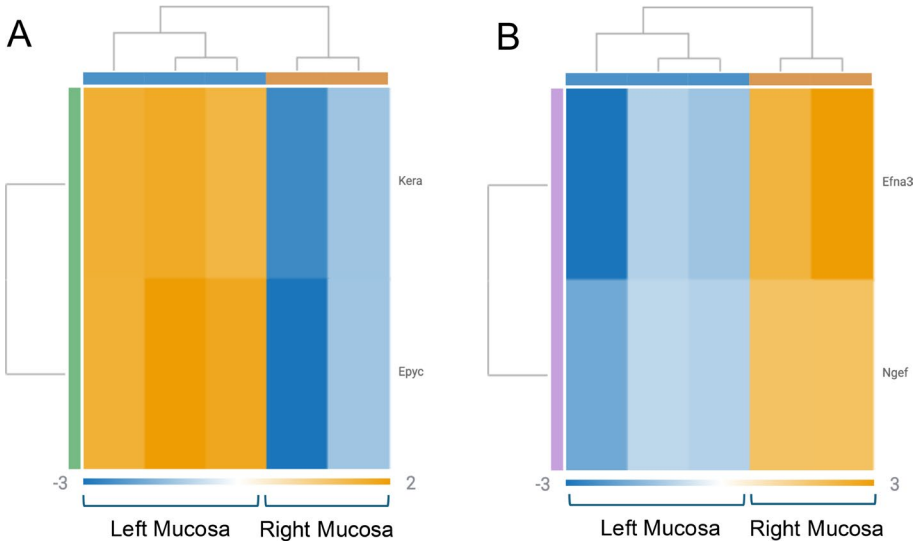


FIGURE 5 | Mucosa pathway heatmaps (FC ≥ 4). (A) Small leucine-rich proteoglycan (SLRP) pathway (e.g., KERA, EPYC upregulated on Left). (B) Ephrin-A signaling (e.g., EFNA3, NGEF downregulated on Left).

4 | Discussion

We investigated whether unilateral denervation produces side-specific transcriptomic responses in the VF MTA and mucosa. Using a primary Left-denervated-versus-Right-denervated comparison, interpreted alongside a separate sham Left-versus-Right analysis of baseline laterality, we identified compartment-specific transcriptional shifts consistent with coordinated remodeling after injury. Because transcription-factor activity was inferred from perturbation signatures rather than measured directly, these findings should be considered hypothesis-generating.

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In MTA, enrichment of endochondral ossification pathways was consistent with extracellular-matrix (ECM) remodeling, with contributing genes including members of the ADAMTS family and SERPINH1. Members of the ADAMTS protease family, notably ADAMTS1/4/5, are canonical versicanases that cleave versican (VCAN) [24], a prominent proteoglycan in the human vocal-fold lamina propria [25], with the potential to reshape provisional matrix and modulate signaling.

Although these mechanisms have primarily been described in the lamina propria, these data identify related transcriptional signatures within VF muscle. SERPINH1 (HSP47) is a collagen-specific chaperone essential for procollagen folding. In rat VF mucosal injury, Serpinh1 knockdown reduced collagen accumulation and reversed fibrosis [26]. Together, these functions align with ECM- and adhesion-related signatures observed here and highlight potential contributors to left-biased remodeling.

In mucosa, Interleukin-4 regulation of apoptosis and TGF-β-dependent ECM pathways suggest immune/matrix reprogramming; ECM/adhesion genes (POSTN, VCAN, HAS2, LUM) predominated and cell-cycle signals (MYC, MCM family) were reduced. At the stricter $|FC| \geq 4$ threshold, SLRP molecules and Ephrin-A forward signaling persisted, implicating proteoglycan composition and cell–cell communication. These results warrant caution: enrichment used combined up/down DEG sets, direction was inferred from gene-level changes and functional impact requires validation. Denervation elicited ECM- and signaling-focused remodeling with distinct patterns in muscle versus mucosa.

In MTA, reduced LIN28 activity was observed in left-denervated tissue relative to right. LIN28 is associated with IGF-2 translational control and myoblast maturation [27]. Reduced NR2F2 (COUP-TFII) activity was also observed; NR2F2 is active in undifferentiated progenitors [28]. NCOA2 also trended lower; NCOA2 is linked to postnatal maturation and fiber-type specification [29, 30].

Increased ESRRA (ERRα) activity was observed in left-denervated MTA relative to right. ESRRA partners with PGC-1α in mitochondrial programs and biogenesis [31–33]. Increased FOXO1 activity was also observed; FOXO1 is associated with skeletal-muscle atrophy and autophagy-related pathways [34, 35]. Conversely, reduced CEBPG (C/EBPγ) activity was observed; C/EBPγ is associated with cellular stress-response networks [36] and autophagy regulation [37].

DUX4 inferred activity was reduced in left-denervated MTA compared to right-denervated MTA. DUX4 is typically silenced in adult skeletal muscle, but can be reactivated under stress or pathological conditions, where it disrupts gene regulation and promotes cell death, most notably in facioscapulohumeral muscular dystrophy (FSHD) [38–41]. Although the role of DUX4 in denervation outside of FSHD is not well-defined, these findings suggest laterality may influence how developmental pathways are engaged in response to neuromuscular injury.

PITX2, a canonical regulator of left–right patterning during development [42], was inferred to be elevated in left-denervated MTA. Given both RLNs were fully transected, this signal likely represents a transcriptional echo of positional identity rather than a functional repair mechanism and may indicate that developmental patterning programs are more strongly re-engaged or retained following left-sided injury, possibly as a stress-related reactivation of embryonic cues. Whether elevated PITX2 in left-denervated MTA contributes to remodeling or simply marks positional memory remains unclear.

MECP2 activity was inferred to be reduced in left-denervated MTA compared to right. MECP2 deficiency reduces α-SMA expression and impairs myofibroblast differentiation and collagen deposition [43], processes critical for structural remodeling. Lower activity of MECP2 on the left suggests diminished capacity to engage these remodeling programs, potentially limiting matrix adaptation compared to the right MTA following denervation.

FOXA1 activity was inferred to be reduced in left-denervated MTA relative to the right. FOXA1 modulates chromatin accessibility and regulates metabolic and structural gene networks [44]. Reduced FOXA1 coincided with asymmetry consistent with broader epigenetic remodeling after left-sided denervation. Increased EGR4 and GATA6 and decreased U2AF1 were observed in left-denervated MTA; these factors are involved in DNA binding [45], hypertrophy [46], and splicing [47], and are noted without mechanistic interpretation.

Epithelial tissues rely on tight regulatory balance between proliferation and differentiation, maintained in part by TFs such as IRF6 [48–51], OVOL1 [52, 53], and OVOL2 [52]. IRF6 governs keratinocyte maturation and epithelial homeostasis [51], while OVOL1/2 promote epithelial identity [52] and adhesion via E-cadherin regulation [54, 55]. In our data, IRF6 activity was inferred to be reduced in left-denervated mucosa compared with right. OVOL1 activity was likewise inferred to be reduced in left-denervated mucosa. OVOL2 activity was inferred to be increased; mixed directions in control conditions limit interpretation.

Increased HES6 activity was inferred in left-denervated mucosa. HES6 is linked to neuronal differentiation and progenitor states. In sensory neuroepithelia like the olfactory epithelium and vomeronasal organ, Hes6 is expressed in globose basal cells—neuronal progenitors—preceding NeuroD during chemosensory neuron differentiation [56]. In contrast, VAX2 activity was decreased; VAX2 overexpression causes ventralization of the developing eye and expansion of the optic stalk, highlighting its role in retinal patterning, with unknown relevance in mucosa [57]. Taken together with reduced IRF6 and OVOL1, these findings indicate attenuation of epithelial stability programs and engagement of regenerative or progenitor-like signals.

Distinct transcriptional responses in MTA and VF mucosa after left-sided RLN denervation underscore tissue-specific remodeling. Left-sided injury engaged pathways linked to epithelial stability, stress adaptation, mitochondrial function, and atrophy. Together, these patterns indicate coordinated remodeling across compartments and raise the possibility of cross-tissue communication. Laterality shaped these programs in a compartment-specific manner: left-biased immune/ECM signatures were

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observed in the mucosa, whereas ossification-related ECM features were noted in the MTA. Bilateral, organ-level frameworks may reconcile these issues: the vocal fold operates as a coupled system in which MTA and mucosa co-determine closure and tissue mechanics. Denervation produces side-dependent responses due to baseline left–right intrinsic laryngeal muscle asymmetry. In our data, left-biased MTA stress/atrophy and mucosal ECM/immune remodeling are consistent with coordinated adaptation rather than isolated tissue responses. Clinically, this asymmetry may narrow recovery windows and should be considered when setting expectations and timing interventions after unilateral RLN injury.

This study has clear constraints: small sample size and sex modeled as a covariate which may obscure sex-specific responses. In addition, TF activity was derived from perturbation signatures rather than direct measurement. Future work should incorporate time-course profiling, cell-type–resolved analyses, and functional tests of key regulators to define mechanisms and assess whether laterality shapes recovery trajectories.

5 | Conclusion

Left RLN denervation yielded side-specific transcriptomic changes in TA and mucosa. Reduced differentiation/stability, increased stress/atrophy, and regenerative/mitochondrial activation were observed in both, with shifts in developmental/epigenetic programs. Findings support coordinated compartmental remodeling and identify laterality as a relevant biological variable. Perturbation-based TF are hypothesis-generating, guiding targeted validation and mechanistic studies of differential recovery.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Processed RNA-sequencing data, metadata, quality control data, TF perturbation outputs, and figure source matrices were deposited in Figshare (DOI: [https://doi.org/10.6084/m9.figshare.31502893] (https://doi.org/10.6084/m9.figshare.31502893)). Raw FASTQ files are available through NCBI Gene Expression Omnibus under accession GSE324707.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** RNA-seq QC and alignment metrics per sample. RNA-seq quality control metrics including sample metadata (condition, side, tissue, RIN) and sequencing/alignment statistics. Q30, Phred quality ≥ 30; % aligned, reads mapped to mRatBN7.2; % duplicated, PCR/optical duplicates; % unique, uniquely mapped reads; % multi-mapped, reads mapping to multiple loci; reads used in counts, HTSeq-assigned gene reads.

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